AI Entities

Text in this color is mainly for understanding.

↳ When creating artificial intelligence, the purpose is to produce entities that can operate independently of human direction.

↳ An entity may act on a human's behalf though.

↳ These entities need to have the following properties:

1. Autonomy
   ↳ Needs no direct human involvement to perform activities.

2. Responsive
   ↳ Must be able to percieve & react to it's environment.

3. Proactivity:
   ↳ Must exhibit goal-directed behaviour

4. Sociability:
   ↳ Able to interact with humans & other Agents.

Agents:

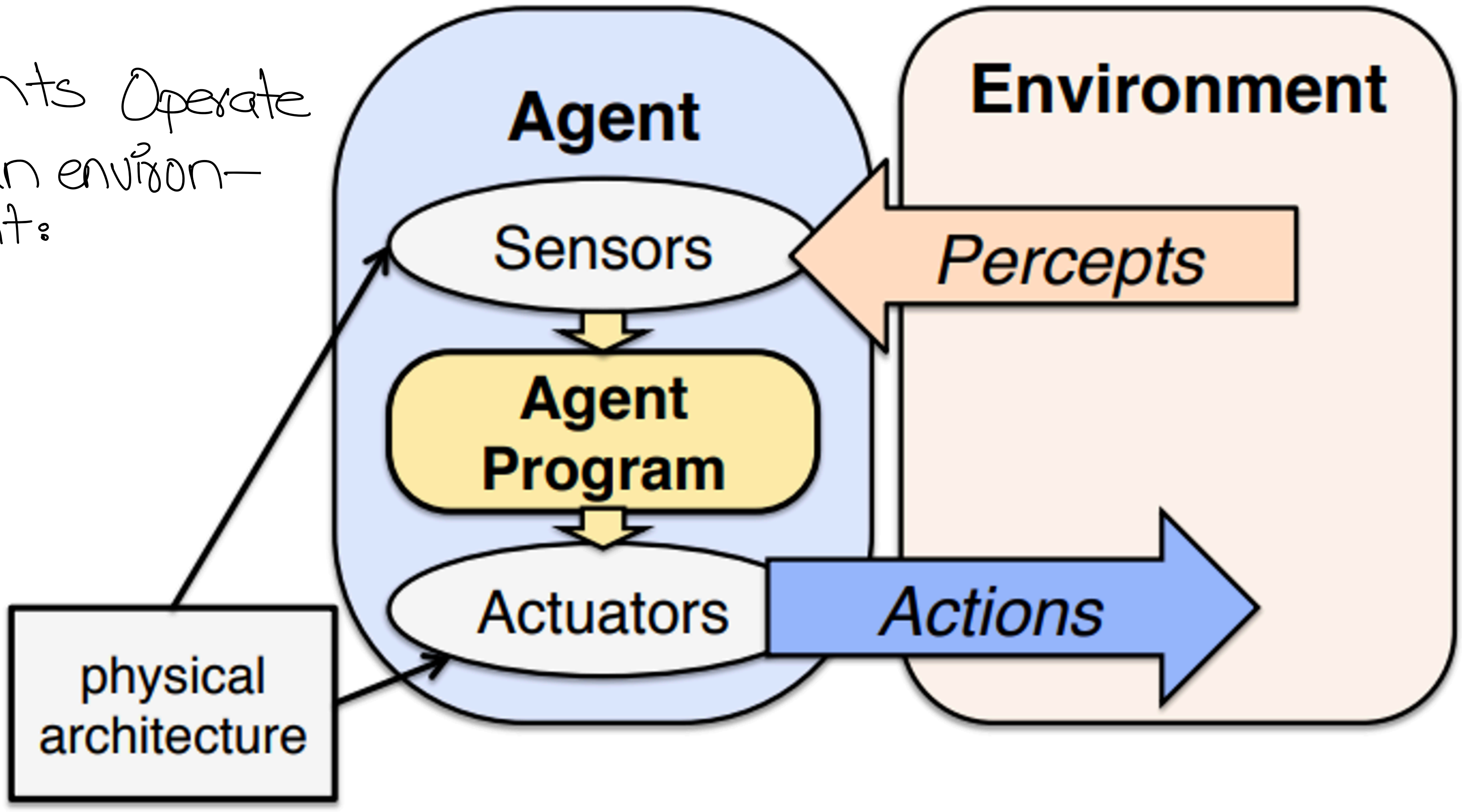
"A software entity that exists in an environment & acts on that environment based on it's preceptions & goals."

↳ The starting points for $\underbrace{\text{agent-based view}}_{\text{Individual perspective}}$ is ourselves

In the sense that we are able to percieve the world (computer vision), performs actions in it (Robotics), and communicate with other people.

We also have knowledge about the world & can use this to draw insights & make decisions. We are also continously learning & adapting.

Agents Operate in an environment:



↳ Agents percieves the world using sensors & acts on it using it's actuators.

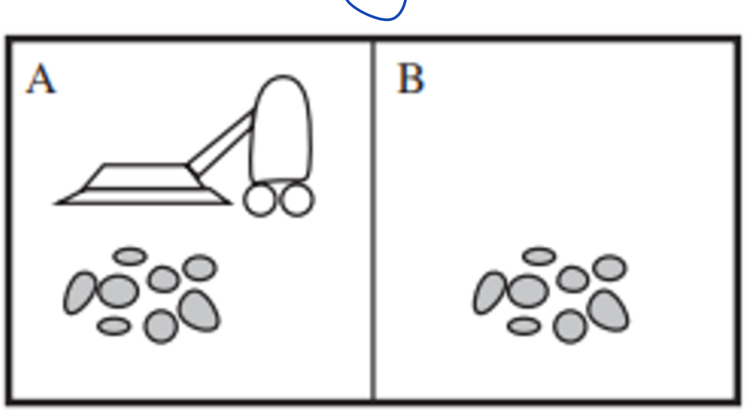
↳ Each of it's components are:

1. Sensor:
   - Eyes, Ears, Nose (humans)
   - Camera, microphone (Agent)

2. Percept:
   - ↳ Perceptual input at any instant
   - ↳ The complete history of what it has so far precieved is called percept sequence.

3. Actuator:
   - Arms, Legs (human/Robot)

Example,
For a Robotic cleaning vaccum (Roomba)
Environment: Rooms
Sensor: A camera
Percepts: Current location, is clean or dirty
Actions: Move left, right or suck



For a self driving car

Environment: Roads

- Challenges: Traffic lights, pedestrians, and weather conditions.

Perception: Camera, radar & lidar

Actions: Steering, acceleration, braking

Perception:

It can be expressed a mapping function

(Sensor) $S \longrightarrow P$ (Perception)

& then

$P \longrightarrow A$ (Action)

Example,

Camera detects red traffic light ⟶ Percieves "red = stop"

Percieves "red = stop" ⟶ Slows down & triggers brakes.

Perception ability:

Non-existence   Omniscient

where,

E: set of percieved states

Includes:

1. Vision: Color, shape

2. Volume: high/low pitches, loud/soft

↳ Two states (s_1 & s_2) are indistinguishable if:

1. They are same $see(s_1) = see(s_2)$

2. The agent has limited perception & can't tell them apart

Agent function:

↳ A complete mapping from sequence of percepts to actions

$$f(<p^{(1)}, p^{(2)}, \ldots, p^{(t)}) = a^{(t)}$$

Agent Program:

↳ It is the implementation of 'f' that runs on a physical architecture.

So, our Agent comprises:

architecture + program

Example,

f("Dust in corner", "Chair in the way", "Battery is low")

=

"Move around the chair, clean the dust in the corner & return to charging dock".

↳ This is a perfect mapping but this not easy because:

- Sensors are imperfect
- Environment changes

↳ So, we approximate the agent function using Agent Progr

Rational Agents:

↳ For each possible percept sequence, a rational agent should select an action that is expected to maximize it's performance measure, given the evidence provided by the percept sequence & built-in knowledge the agent has.

↳ Rationality depends on four things:

1. Performance measure that defines sucess
2. Agents prior knowledge of environment
3. Actions that agent can perform
4. Agent's percept sequence to date

Example,

For self-driving car,

1. Safely reach the destination while minimizing travel time. (Performance measure)
2. Traffic laws, road maps, car's capabilities (Prior knowledge / Environment)
3. Objects detected by sensors. (Percept sequence/ sensors)
4. Accelerate, brake, change lanes. (Possible actions/ actuators)

↳ The car agent continously percieves it's surroundings, considers actions like changing lanes to overtake a slow car, and selects the action that alligns best with it's goals, considering safety & efficiency.

Rationality Vs Omniscience

↳ Rational agent chooses the best possible agent given what it knows.

↳ An omniscient agent would know everything about the world.

↳ Rational is different from being perfect

Rational ↓ maximizes expected output

perfect ↓ maximizes actual output

Specifying the task environment

↳ Task environments are essentially the problems to which agents are the solution.

↳ These contain everything needs to know & interact with in order to act rationally.

PEAS (Performance measure, Environment, Actuators, sensors)

Autonomy in Agents

↳ The autonomy of an agent is the extent to which its behaviour is determined by it's own experience, rather than knowledge of designer.

↳ It's actions must not be fully determined by the designer.
↳ It's actions must not be fully random i.e it must have a clear goal.
↳ Learns from experience.

Environment types:

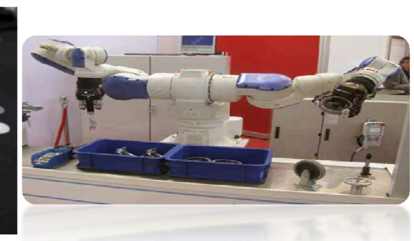
1. Fully observable vs partially observable

↳ If an agents' sensors give it access to the complete state of the environment at each point in time, then the task-environment is fully observable i.e sensors detect all aspects that are relevant to the task at hand.

↳ An environment might be partially observable due to noisy & inaccurate sensors.
Example,
An automated taxi can't see what other drivers are thinking.

↳ If agent has no sensors, then the environment is unobservable.

Poker Game (Partially)



Part Picking Robot (Fully)

if it deals with probabilities — If possibilities listed out without quantification

2. Deterministic vs stochastic OR non-deterministic

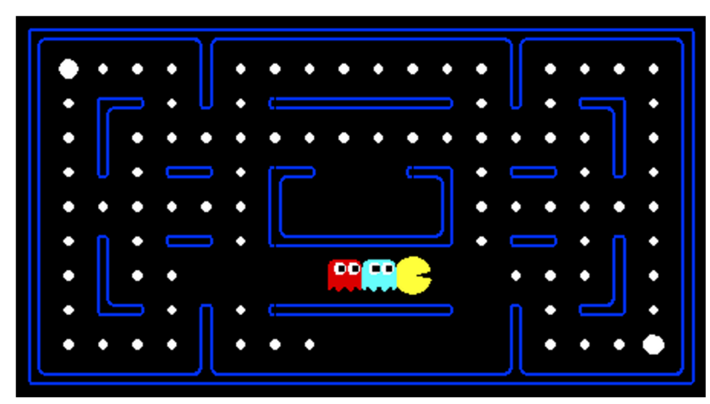
↳ Is the next state of the environment completely determined by the current state & the agent's action?
↳ If an environment is partially observable, then it could appear non-deterministic.
↳ For instance, taxi driving is non-deterministic as we can never predict what will happen next on the road. Tire may blow or engine may sieze up. The vaccum world is deterministic with some non-deterministic elements such as randomly appearing dust

3. Episodic vs Sequential

- Episodic:
  ↳ Agent recieves a single percept per episode. It's action does not affect future percepts
  Example,
  classification

- Sequential:
  ↳ Agent recieves a sequence of percepts. The current action affects future percepts
  Example,
  taxi driving, chess playing

VS.



4. Static vs dynamic

Is the world changing while agent is thinking?

- Static:
  ↳ Environment doesn't change unless agent performs an action
  Example,
  cross world puzzle

- Dynamic:
  ↳ Environment changes even when the agent doesn't do anything?
  Example,
  Road

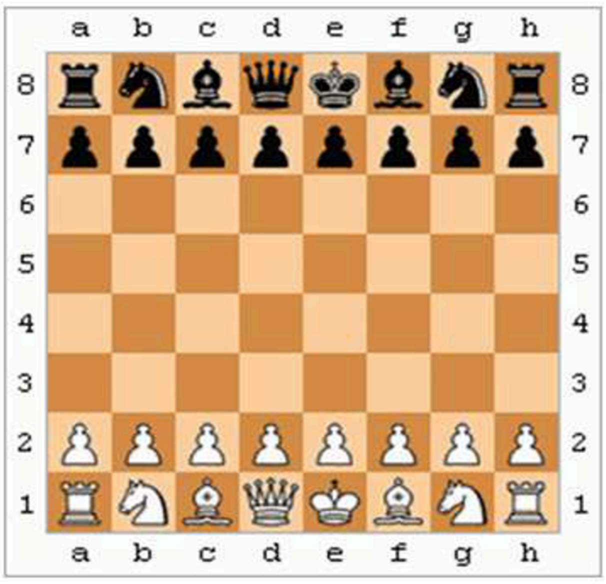
- Semi-dynamic:
  ↳ The environment is static but the agent's per-

formance score changes.
Example,
chess played with a clock

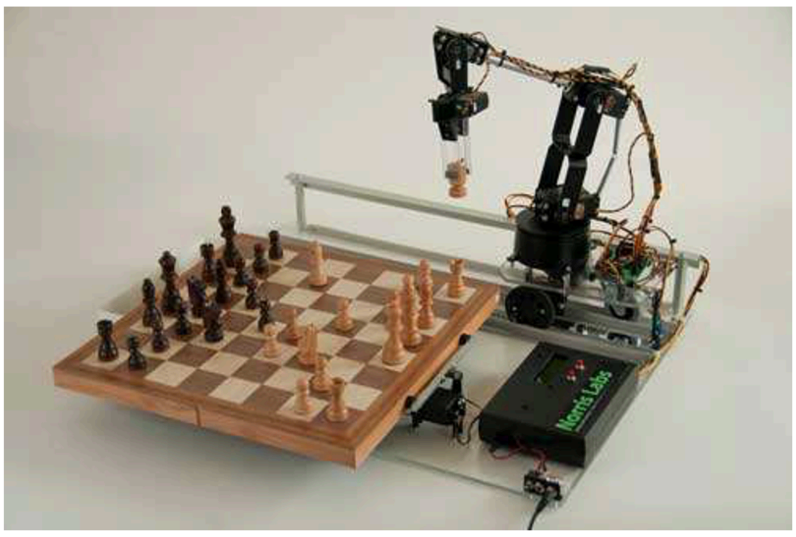
5. Discrete VS continuous:

- Continuous:
  ↳ Time, percepts, and actions are continous
  Example,
  Driving a car

- Discrete:
  ↳ Time, percepts and actions are discrete
  Example,
  Playing a board game.



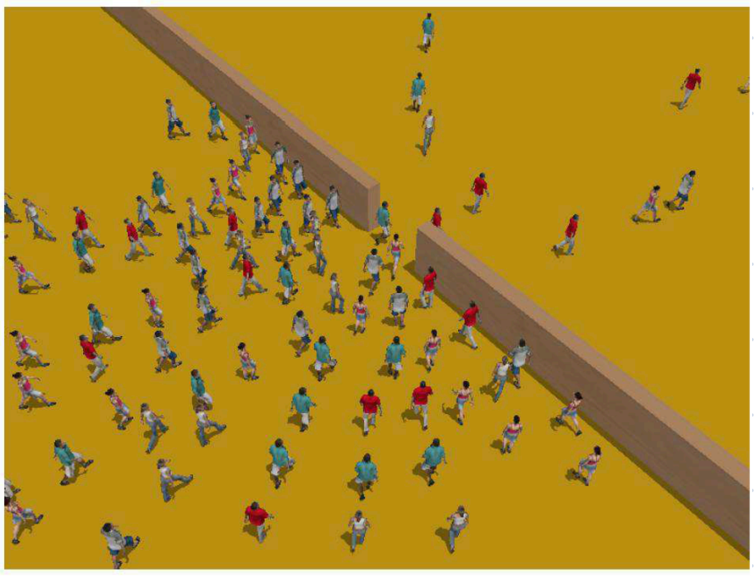
VS.



5. Single agent VS multi-agent

↳ Is an agent operating by itself in the environment?

VS.

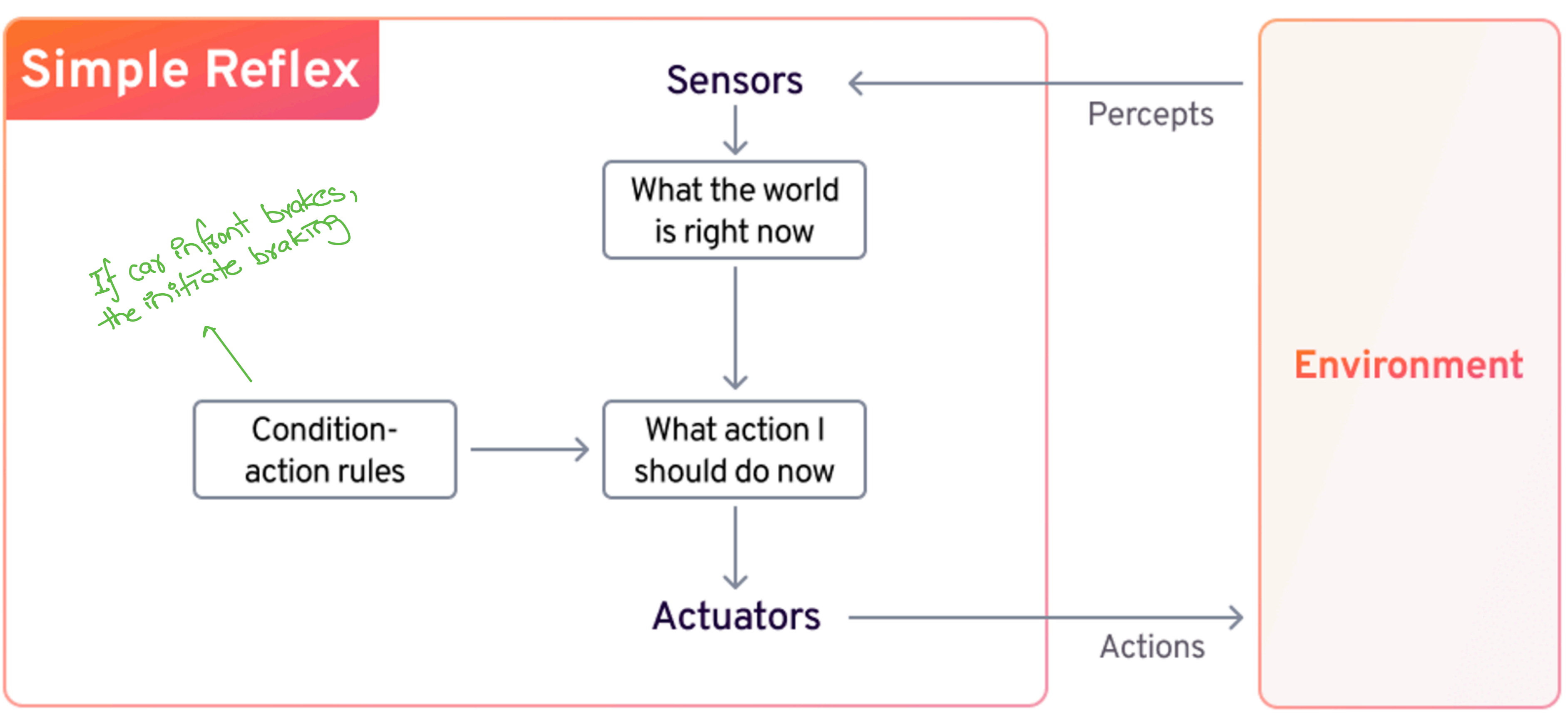


Summary

	Observable	Deterministic	Episodic	Static	Discrete	Agents
Cross Word	Fully	Deterministic	Sequential	Static	Discrete	Single
Poker	Fully	Stochastic	Sequential	Static	Discrete	Multi
Backgammon	Partially	Stochastic	Sequential	Static	Discrete	Multi
Taxi driver	Partially	Stochastic	Sequential	Dynamic	Conti	Multi
Part picking robot	Partially	Stochastic	Episodic	Dynamic	Conti	Single
Image analysis	Fully	Deterministic	Episodic	Semi	Conti	Single

Agent types

1. Simple reflex based models



- Simple but very limited intelligence
- Doesnot maintain an internal state
- Action depend only on current precept.

function SIMPLE-REFLEX-AGENT(*percept*) **returns** an action
persistent: *rules*, a set of condition–action rules

state ← INTERPRET-INPUT(*percept*)
rule ← RULE-MATCH(*state*, *rules*)
action ← *rule*.ACTION
return *action*

2. Model-based reflex agent

↳ Maintain an internal state about the world

↳ Updating this internal state requires two types of knowledge to be encoded in the agent program.

1. How the world changes over time?

a. Effects of agent's actions on the environment

b. How world evolves independent of the agent

Example,

Agent turns steering wheel clockwise then car turns right

Raining outside then car's cameras get wet.

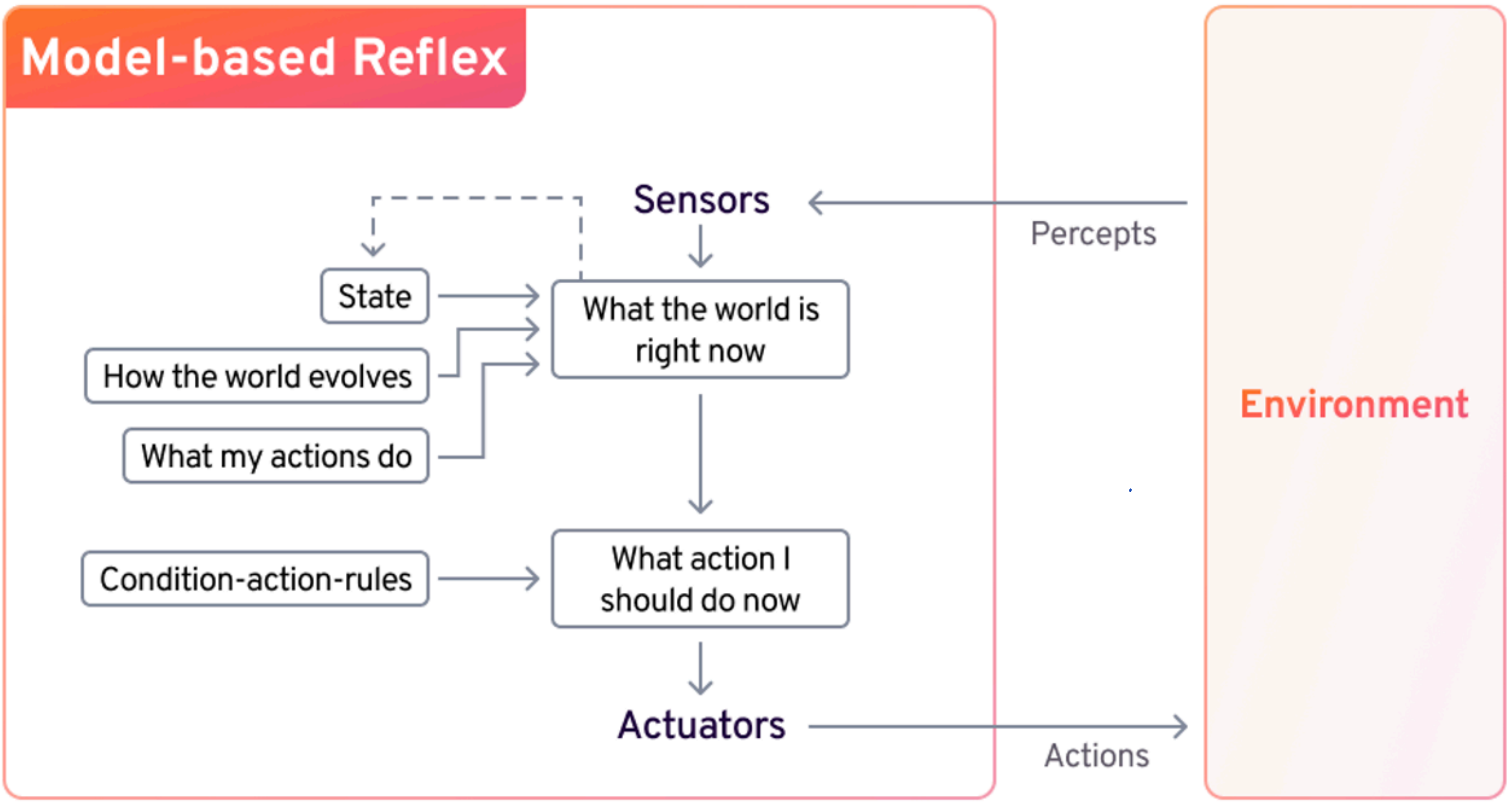
↳ This knowledge about how the world works is called transition model.

2. We need information about how the state of the world is reflected in the agent's percept.

For example,

when the car in front initiates braking, one or more illuminated red agents appear in the forward-facing camera image

when the camera gets wet, droplet-shaped objects appear in the image



function MODEL-BASED-REFLEX-AGENT(*percept*) **returns** an action
persistent: *state*, the agent's current conception of the world state
transition_model, a description of how the next state depends on the current state and action
sensor_model, a description of how the current world state is reflected in the agent's percepts
rules, a set of condition–action rules
action, the most recent action, initially none

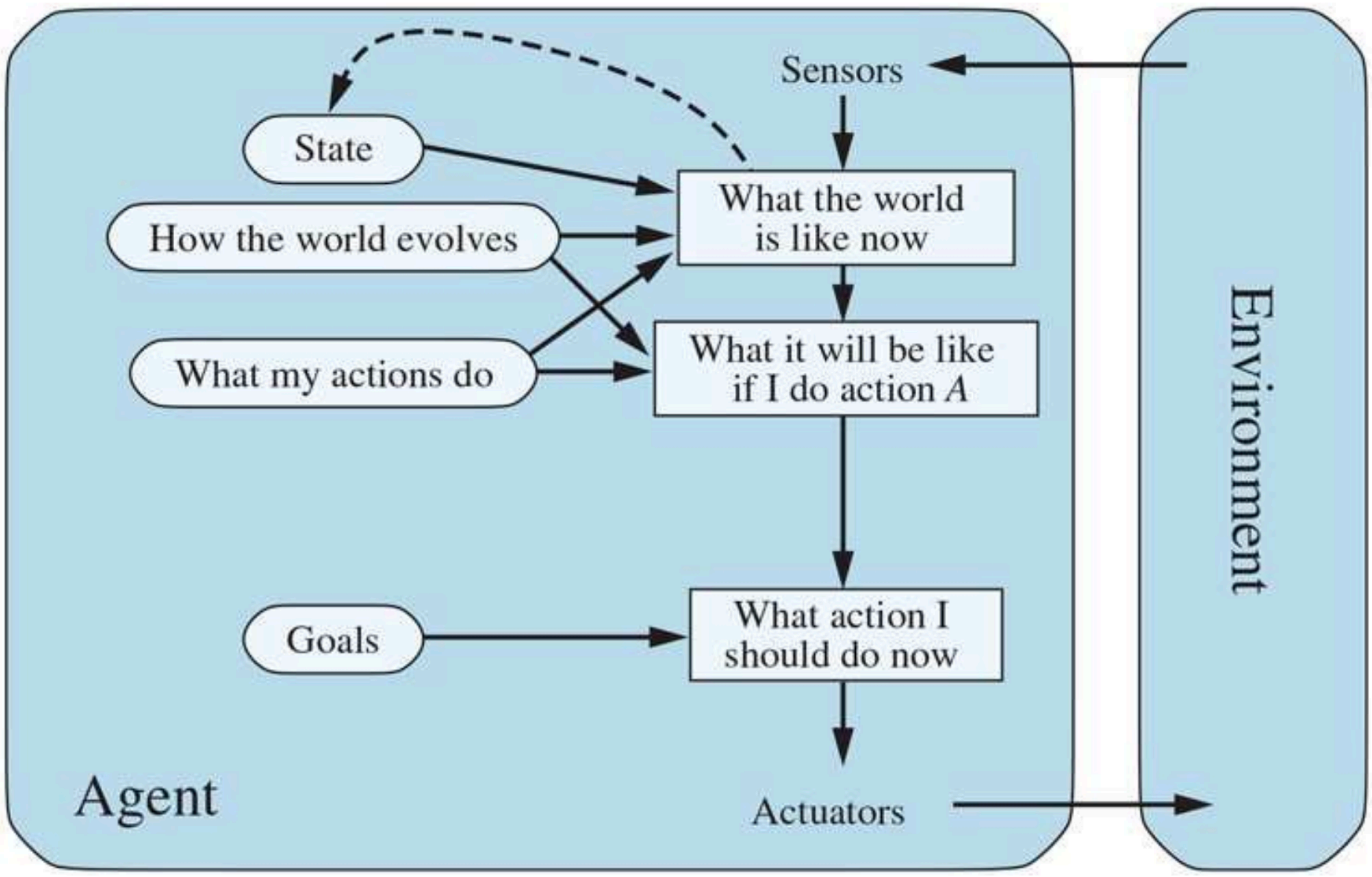
state ← UPDATE-STATE(*state*, *action*, *percept*, *transition_model*, *sensor_model*)
rule ← RULE-MATCH(*state*, *rules*)
action ← *rule*.ACTION
return *action*

3. Goal-based Agents

↳ Further, we need to decide what to do & that is based on what our goal is.

For example,
A taxi can turn left, right or go straight but the correction decision would depend on it's destination.

↳ Search & planning



"the quality of being usefull"

4. Utility-based agents

↳ Goals are not enough to generate high-quality behaviour in most environments.

↳ Agent may have conflicting goals → Reach safely vs Reach fast

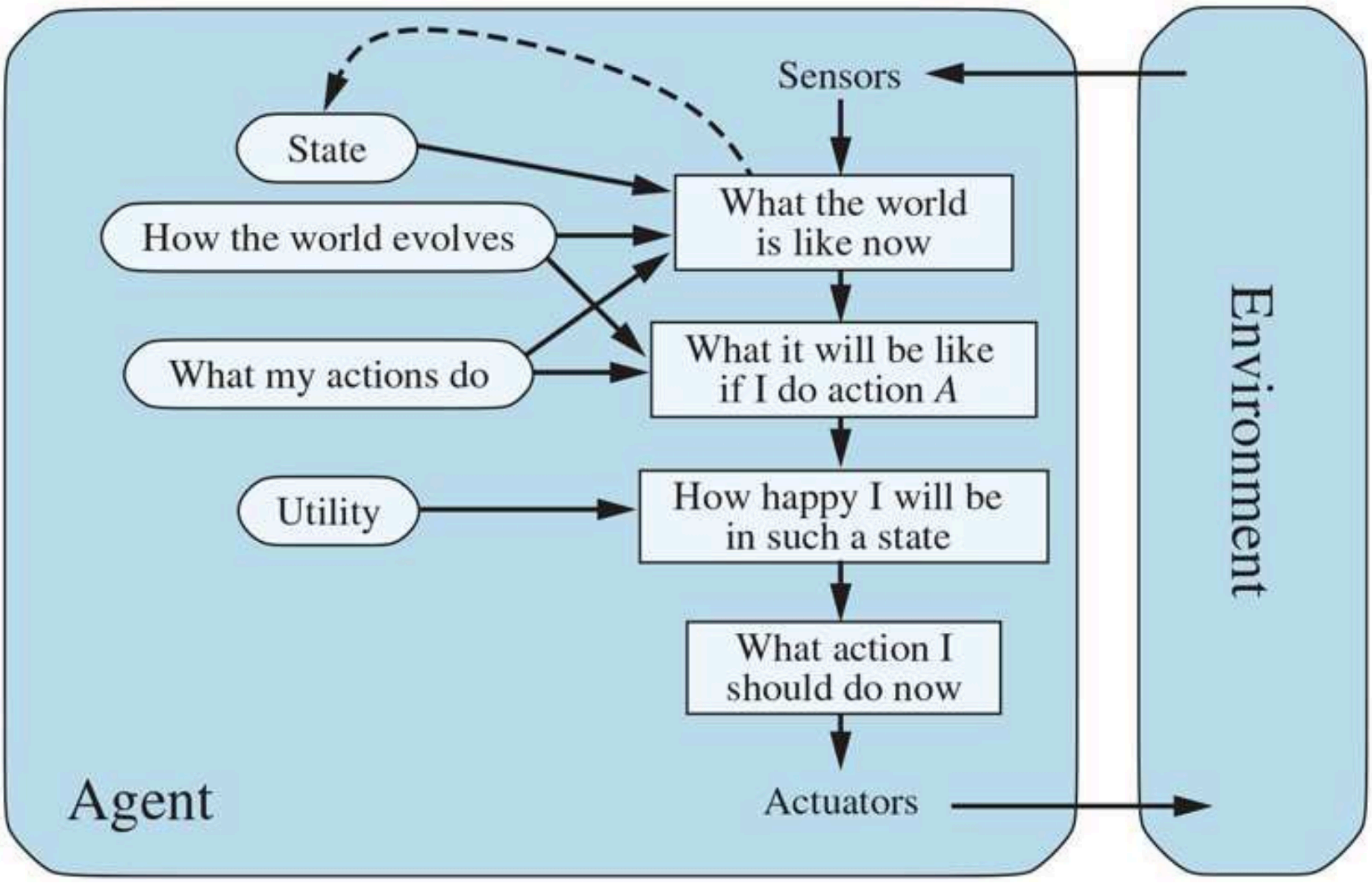
Think as agent's internal understanding of Good v bad actions

↳ So, Agent has a utility function, which (ideally) approximates the external performance measure.

↳ Furthermore, goals are inadequate but such agent can still make rational decisions.
For instance,

1. Conflicting goals (speed vs safety) the utility function provides an optimal trade-off.

2. Many goals & none of which can achieved with certainty, utility provides a way in which the likelihood of success can be weighted against the importance of goals.



5. Learning agents

- Allows the agent to operate in initially unknown environments & to become more competent than it's initial knowledge alone might allow.

- A learning agent can be divided into four conceptual components.

  1. Learning element → Responsible for making improvements

  2. Performance element → responsible for selecting external actions. (Entire agent)

  3. Critic → Responsible for judging how the agent is doing & determines how the performance element should be modified.

     - Operates on basis of a fixed performance standard

  4. Problem generator → Responsible for suggesting

actions that will lead to
new & informative experiences

For example,
If an automated taxi brakes on a wet road, then it will learn how much deceleration is achieved & whether it skids off the road.